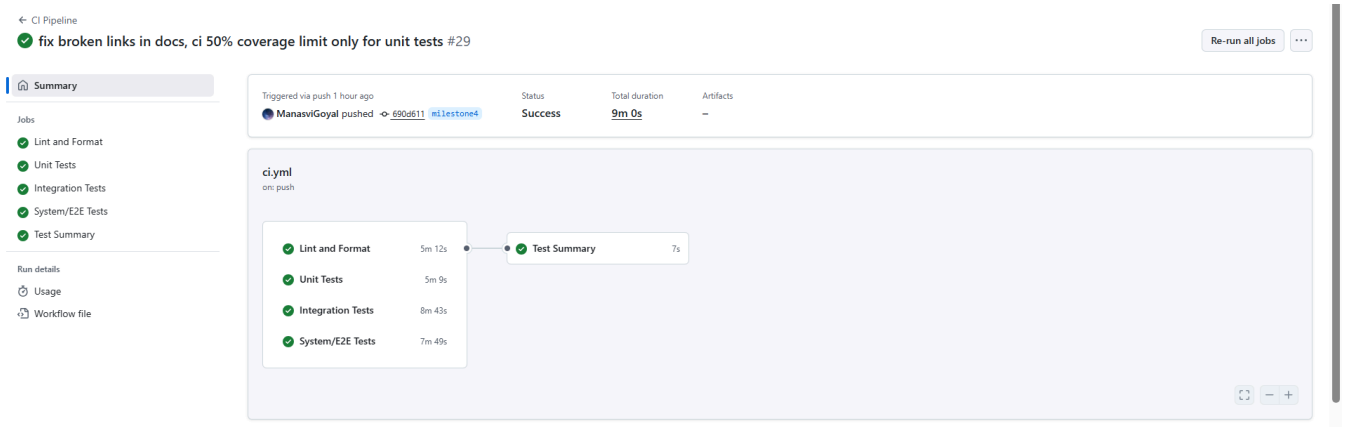
Milestone 4: CI Evidence

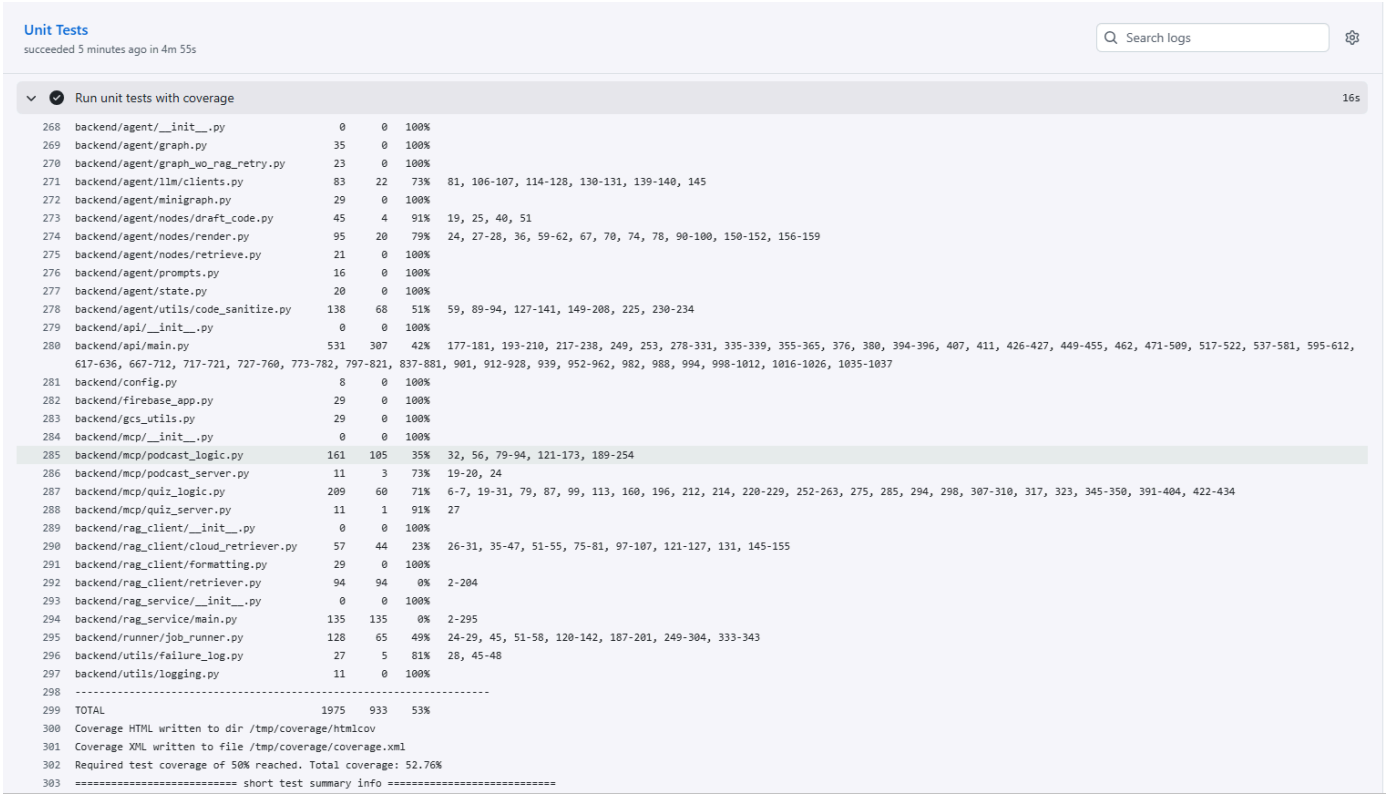
1. Passing CI run

- Successful build and linting
- All tests (Unit, Integration, System/E2E) passing



2. Code coverage reports

a) Unit Tests (53%)




```
backend/agent/__init__.py                 0      0   100%
backend/agent/graph.py                   35      0   100%
backend/agent/graph_wo_rag_retry.py      23      0   100%
backend/agent/llm/clients.py             83     22    73%   81, 106-107, 114-128, 130-131, 139-140, 145
backend/agent/minigraph.py               29      0   100%
backend/agent/nodes/draft_code.py        45      4    91%   19, 25, 40, 51
backend/agent/nodes/render.py            95     20    79%   24, 27-28, 36, 59-62, 67, 70, 74, 78, 90-100, 150-152, 156-159
backend/agent/nodes/retrieve.py          21      0   100%
backend/agent/prompts.py                 16      0   100%
backend/agent/state.py                   20      0   100%
backend/agent/utils/code_sanitize.py    138     68    51%   59, 89-94, 127-141, 149-208, 225, 230-234
backend/api/__init__.py                   0      0   100%
backend/api/main.py                     531    307    42%   177-181, 193-210, 217-238, 249, 253, 278-331, 335-339, 355-365, 376, 380, 394-396, 407, 411, 426-427, 449-455, 462, 471-509, 517-522, 537-581, 595-612,
617-636, 667-712, 717-721, 727-760, 773-782, 797-821, 837-881, 901, 912-928, 939, 952-962, 982, 988, 994, 998-1012, 1016-1026, 1035-1037
backend/config.py                         8      0   100%
backend/firebase_app.py                  29      0   100%
backend/gcs_utils.py                     29      0   100%
backend/mcp/__init__.py                   0      0   100%
backend/mcp/podcast_logic.py            161    105    35%   32, 56, 79-94, 121-173, 189-254
backend/mcp/podcast_server.py            11      3    73%   19-20, 24
backend/mcp/quiz_logic.py               209     60    71%   6-7, 19-31, 79, 87, 99, 113, 160, 196, 212, 214, 220-229, 252-263, 275, 285, 294, 298, 307-310, 317, 323, 345-350, 391-404, 422-434
backend/mcp/quiz_server.py               11      1    91%   27
backend/rag_client/__init__.py            0      0   100%
backend/rag_client/cloud_retriever.py    57     44    23%   26-31, 35-47, 51-55, 75-81, 97-107, 121-127, 131, 145-155
backend/rag_client/formatting.py         29      0   100%
backend/rag_client/retriever.py          94     94     0%   2-204
backend/rag_service/__init__.py           0      0   100%
backend/rag_service/main.py             135    135     0%   2-295
backend/runner/job_runner.py            128     65    49%   24-29, 45, 51-58, 120-142, 187-201, 249-304, 333-343
backend/utils/failure_log.py             27      5    81%   28, 45-48
backend/utils/logging.py                 11      0   100%
-----------------------------------------------------------------------
TOTAL                                  1975    933    53%
Coverage HTML written to dir /tmp/coverage/htmlcov
Coverage XML written to file /tmp/coverage/coverage.xml
Required test coverage of 50% reached. Total coverage: 52.76%
========================= short test summary info ==========================
```

b) Integration Tests (46%)

Integration Tests
succeeded 3 minutes ago in 8m 10s

Search logs

Run integration tests 3m 38s

```
backend/__init__.py                      0      0   100%
backend/agent/__init__.py                0      0   100%
backend/agent/graph.py                  35      1    97%   75
backend/agent/graph_wo_rag_retry.py     23     23     0%   2-41
backend/agent/llm/clients.py            83     45    46%   46-47, 50, 52-53, 81, 106-107, 110-131, 135-148, 150-151, 173
backend/agent/minigraph.py              29      1    97%   54
backend/agent/nodes/draft_code.py       45      7    84%   16, 18, 25, 28, 33, 40, 51
backend/agent/nodes/render.py           95     28    71%   20, 25-29, 41-44, 64-67, 70, 74, 80, 90-100, 127-130, 150-152, 156-159
backend/agent/nodes/retrieve.py         21      5    76%   25-26, 40-42
backend/agent/prompts.py                16      3    81%   89-90, 99
backend/agent/state.py                  20      0   100%
backend/agent/utils/code_sanitize.py   138     73    47%   16, 32, 35, 51-52, 59, 89-94, 127-141, 149-208, 225, 230-234
backend/api/__init__.py                  0      0   100%
backend/api/main.py                    531    296    44%   27-40, 193-210, 217-238, 278-331, 335-339, 429, 454, 462, 471-509, 517-522, 537-581, 595-612, 617-636, 667-712, 717-721, 727-760, 773-782, 797-821, 837-
881, 901, 912-928, 939, 952-962, 982, 988, 994, 998-1012, 1016-1026, 1035-1037
backend/config.py                        8      0   100%
backend/firebase_app.py                 29     19    34%   26-46, 50
backend/gcs_utils.py                    29     16    45%   23-25, 30-34, 39-41, 55-57, 62-64
backend/mcp/__init__.py                  0      0   100%
backend/mcp/podcast_logic.py           161     54    66%   24-28, 49-50, 54, 56, 68, 79-94, 123, 139-155, 166, 196, 218-228, 240-241, 247, 251-252
backend/mcp/podcast_server.py           11      3    73%   19-20, 24
backend/mcp/quiz_logic.py              209    133    36%   6-7, 27-31, 38-44, 64-119, 131-161, 168-189, 193-197, 202-231, 236-275, 285, 288, 294, 298, 307-310, 313, 317, 320, 323, 335
backend/mcp/quiz_server.py              11      1    91%   27
backend/rag_client/__init__.py           0      0   100%
backend/rag_client/cloud_retriever.py   57     26    54%   31, 40-41, 43, 45, 51-55, 75-81, 98, 107, 121-127, 131, 152-155
backend/rag_client/formatting.py        29     27     7%   6-32
backend/rag_client/retriever.py         94     94     0%   2-204
backend/rag_service/__init__.py          0      0   100%
backend/rag_service/main.py            135    135     0%   2-295
backend/runner/job_runner.py           128     65    49%   24-29, 45, 51-58, 120-142, 187-201, 249-304, 333-343
backend/utils/failure_log.py            27      9    67%   39-48
backend/utils/logging.py                11      0   100%
-----------------------------------------------------------------------
TOTAL                                 1975   1064    46%
Coverage HTML written to dir htmlcov
Coverage XML written to file coverage.xml
================= 240 passed, 1 warning in 214.39s (0:03:34) =================
```

c) System E2E Tests (45%)

System/E2E Tests
succeeded 2 minutes ago in 7m 49s

Search logs

Run system tests 2m 54s

```
backend/__init__.py                      0      0   100%
backend/agent/__init__.py                0      0   100%
backend/agent/graph.py                  35      1    97%   75
backend/agent/graph_wo_rag_retry.py     23     23     0%   2-41
backend/agent/llm/clients.py            83     45    46%   46-47, 50, 52-53, 81, 106-107, 110-131, 135-148, 150-151, 173
backend/agent/minigraph.py              29      1    97%   54
backend/agent/nodes/draft_code.py       45      7    84%   16, 18, 25, 28, 33, 40, 51
backend/agent/nodes/render.py           95     28    71%   20, 25-29, 41-44, 64-67, 70, 74, 80, 90-100, 127-130, 150-152, 156-159
backend/agent/nodes/retrieve.py         21      5    76%   25-26, 40-42
backend/agent/prompts.py                16      3    81%   89-90, 99
backend/agent/state.py                  20      0   100%
backend/agent/utils/code_sanitize.py   138     73    47%   16, 32, 35, 51-52, 59, 89-94, 127-141, 149-208, 225, 230-234
backend/api/__init__.py                  0      0   100%
backend/api/main.py                    531    304    43%   27-40, 193-210, 217-238, 278-331, 335-339, 394-396, 429, 449-455, 462, 471-509, 517-522, 537-581, 595-612, 617-636, 667-712, 717-721, 727-760, 773-782,
797-821, 837-881, 901, 912-928, 939, 952-962, 982, 988, 994, 998-1012, 1016-1026, 1035-1037
backend/config.py                        8      0   100%
backend/firebase_app.py                 29     19    34%   26-46, 50
backend/gcs_utils.py                    29     16    45%   23-25, 30-34, 39-41, 55-57, 62-64
backend/mcp/__init__.py                  0      0   100%
backend/mcp/podcast_logic.py           161     55    66%   22, 24-28, 49-50, 54, 56, 68, 79-94, 123, 139-155, 166, 196, 218-228, 240-241, 247, 251-252
backend/mcp/podcast_server.py           11     11     0%   1-24
backend/mcp/quiz_logic.py              209    134    36%   6-7, 24, 27-31, 38-44, 64-119, 131-161, 168-189, 193-197, 202-231, 236-275, 285, 288, 294, 298, 307-310, 313, 317, 320, 323, 335
backend/mcp/quiz_server.py              11     11     0%   1-27
backend/rag_client/__init__.py           0      0   100%
backend/rag_client/cloud_retriever.py   57     26    54%   31, 40-41, 43, 45, 51-55, 75-81, 98, 107, 121-127, 131, 152-155
backend/rag_client/formatting.py        29     27     7%   6-32
backend/rag_client/retriever.py         94     94     0%   2-204
backend/rag_service/__init__.py          0      0   100%
backend/rag_service/main.py            135    135     0%   2-295
backend/runner/job_runner.py           128     65    49%   24-29, 45, 51-58, 120-142, 187-201, 249-304, 333-343
backend/utils/failure_log.py            27      9    67%   39-48
backend/utils/logging.py                11      0   100%
-----------------------------------------------------------------------
TOTAL                                 1975   1092    45%
Coverage HTML written to dir htmlcov
Coverage XML written to file coverage.xml
================= 174 passed, 3 warnings in 171.28s (0:02:51) =================
```